Team members' names:

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Semester Project Planning Document

Purpose of Analysis Section

a. What is the problem you want to study (i.e., your curiosity)?

Ans) The problem we want to study is whether Virginia Tech's campus infrastructure, including transportation systems, dining services, classrooms, and student facilities, can adequately accommodate the growing student population. As enrollment increases, students may experience overcrowding, longer wait times, and difficulty accessing important campus resources. This study aims to identify where these challenges occur and how they affect students' daily experiences on campus.

b. What questions would you need to answer to provide information that would help solve the problem you stated in a.?

Ans) In order to address this problem, it is necessary to consider the different ways in which students interact with different aspects of Virginia Tech's infrastructure to determine where the pressure points exist. This may include understanding the different ways in which students commute to university, the modes of transportation that students utilize the most, as well as the level of satisfaction that students have when it comes to BT Transit and parking services provided by the university. It may also be necessary to consider the different ways in which the university's facilities, such as the dining hall, library, gym, as well as lecture rooms, are utilized by the student population, the level of frequency that students utilize such facilities, as well as the level of time that students spend in line for services, among other factors. It may also be necessary to consider whether students face challenges when it comes to registering for classes, as well as the utilization of different digital services such as Canvas and Hokie SPA, among other factors in order to determine the different areas of the university's infrastructure that may be impacted by the growing student population.

c. What answers would you need to give your audience(s) to help them make decisions to solve the problem in a.?

Ans) In order to help them address this issue, the audience would need to gain clear insights into the specific areas of campus infrastructure that are experiencing the greatest levels of strain, such as the specific services that are experiencing the greatest levels of overcrowding, at what time these students are facing these difficulties, and the specific resources that students are having the greatest difficulty accessing. Additionally, the analysis should provide students with clear insights into how the reliability of campus transportation, the wait times for dining services, the availability of classrooms, and the availability of campus facilities impact students' overall routines and satisfaction levels

Write down the justification/motivation for your analysis (i.e., the who and the what).

a. Who wants to know the answers to your analysis (your audience)?

Ans) The target audience for this analysis will include Virginia Tech administrators, planning teams, dining services management, and BT Transit management. These groups are responsible for managing resources on campus and ensuring that resources can effectively meet the needs of students on campus. The results of this analysis can be used by these groups to better understand areas on campus where students are having issues with transportation, dining availability, classroom space, and availability of campus resources. This can include areas such as overcrowding, waiting periods, and availability of services. By doing so, the university can better make decisions on how to improve services and resources for students on campus.

b. What is it you need your audience to know and do?

Ans) Our audience needs to understand which campus services and facilities experience the highest demand and where students face the most inefficiencies in their daily routines. This includes identifying patterns such as peak usage times, common transportation challenges, and areas where students experience delays or overcrowding. With this information, administrators and campus service managers can take targeted actions such as adjusting service capacity during peak hours, improving transportation scheduling, expanding dining options, or optimizing the use of campus facilities. Our goal is to help decision makers prioritize improvements and implement solutions that make campus services more efficient, accessible, and supportive of the student population.

c. Practical implications – how will people be able to use your analysis?

Ans) The findings of this analysis may be useful to university administrators and service managers in better understanding how students utilize infrastructure within the university and what improvements may be needed to meet student demand. In other words, by analyzing student patterns in satisfaction and demand for infrastructure, it can be useful in gaining a better understanding of how well students are being supported with the existing infrastructure. In effect, it may be useful in making better decisions to meet the needs of a growing student body with regard to infrastructure within the university.

d. Can you include an anecdotal observation that explains the phenomena you are examining?

Ans) Some members of our group have personally experienced the effects of crowded campus services. For example, a few of us work in Virginia Tech Dining and receive meal passes at the end of our shifts. When we try to use these meal passes at dining halls such as Perry's, Turner, or Owens, we often encounter very long lines and significant wait times, especially during peak meal hours. Some of us also frequently use library study rooms for group work and have noticed that these spaces are often fully booked. If we try to reserve a room the same day or even the day before, it is usually unavailable, which forces us to book several days in advance. Experiences like these illustrate how high demand for campus services can create congestion and make it difficult for students to access resources when they need them.

Goals of your Analysis Section

a. What actionable outcomes do you hope to achieve through your analysis?

Ans) Through this analysis, we hope to first, gain an insight into the current state of Virginia Tech's infrastructure, then use that knowledge to test it against the strength in numbers (of students) and realize the effectiveness of the services and buildings. Finally, we wish to hypothesize if the university's facilities are truly enough to support the needs of the growing student population. This project was undertaken to understand the extent to which Virginia Tech needs to adapt, develop or change it's resources and offerings.

b. What types of findings (answers) do you think you will need to convince others of the validity of your findings, the interpretation of the findings, and your suggested actionable outcomes?

Ans) The most important part of any project is the data we collect to understand the key question. As is with this project, we will resort to real-time collected data to prove our point. From that, we will focus on the answers that students have given regarding ratings, experiences and opinions of the facilities because we believe that when the facts come directly from the source (i.e., the students), the findings are not only more valid but also more accurate. Especially any open-ended questions, triangulated data, that allow students to freely express their opinions, will be used to match the students' ratings with their answers - giving us added support to their answers. Furthermore, we will focus on recurring themes on issues mentioned consistently across different student demographics to prove that our suggested outcomes are not just reactions to outliers, but necessary solutions for the majority.

Objectives of Analysis (Methods)

a. What data do you need to perform your analysis and create your data story?

Ans) We collected survey data from Virginia Tech students about their campus experience including transportation, dining, infrastructure, digital services, and campus resources. The dataset includes demographic and academic information such as student status, college major, and the number of in-person classes taken per week. It also captures behavioral data like how students commute to campus, the time they typically arrive, and how long it takes them to get to class. Additionally the survey gathers feedback on campus services, including wait times at dining halls, satisfaction with BT Transit, and ratings of online systems like Canvas, Hokie SPA, and eduroam.

b. What sources did you use to guide your data collection?

Ans) To come up with our survey questions, we drew from a mix of our own campus experiences and several sources we researched along the way. Since some of us work in VT Dining and regularly deal with long lines during peak hours, that directly shaped the questions we asked about dining wait times and bottlenecks. For the commuting and transit section, we looked at the Virginia Tech Transportation Services website (parking.vt.edu) and the Blacksburg Transit site (ridebt.org) to better understand what services are actually available to students and what would be worth asking about. Our digital infrastructure questions came from reviewing the VT IT Strategic Plan (it.vt.edu), which gave us a clearer picture of the platforms students rely on day to day. We also referenced the Virginia Tech Sustainability Annual Report 2024-2025 and the Board of Visitors Residential and Athletic Infrastructure Planning report from February 2026, which helped us frame the bigger idea of whether campus infrastructure is keeping up with student growth. On top of all that, we used standard Likert-scale and matrix question formats because they made the most sense for capturing satisfaction levels and how often students use different services.

c. What steps did you take to integrate your data?

Ans) To integrate the data we initially collected all responses through a google form and exported the dataset into an Excel spreadsheet. The exported file served as our central dataset where all survey responses were stored in a structured tabular format. After organizing the spreadsheet, we imported the Excel file into Tableau and connected it as the primary data source. This allowed us to map variables, create relationships between fields, and prepare the data for visualization and analysis in dashboards.

d. What steps did you take to clean your data?

Ans) Before uploading the dataset to Tableau, we performed multiple data cleaning steps in Excel to ensure the data was consistent and usable. We standardized column names, formatted timestamps and time related fields, and ensured numerical responses were correctly recognized as numeric values. We also checked for missing, incomplete, or inconsistent responses and corrected formatting issues in categorical fields such as transportation methods or satisfaction ratings. These steps helped ensure the dataset was accurate and structured properly for analysis and visualization.

e. Describe your statistical analysis. What did you do, how did you organize it, and what did your testing reveal?

Ans) We organized the survey data into categories such as transportation, dining, digital services, and campus infrastructure. Using Tableau, we created visualizations and calculated basic statistics like averages, counts, and percentages to identify trends. We analyzed patterns such as common transportation methods, average commute times, and satisfaction ratings for services like BT Transit, dining halls, and campus digital platforms. The analysis revealed that students across all colleges experience infrastructure stress, with dining wait times averaging 16.1 minutes, gym crowding peaking between 5 and 8 PM, and digital platforms like Hokie SPA consistently receiving the lowest satisfaction scores. Overall infrastructure stress averaged 3.505 out of 5 across all respondents, confirming that the strain is widespread and not limited to any one service or student group.

f. What artifacts from your analysis are important to carry forward to tell your story?

Ans) The most important artifacts from our analysis are the Tableau dashboards and summary visualizations that clearly communicate the patterns we found in the survey data. These include a treemap showing foot traffic stress by academic building, a heat map of dining location peak times, a box plot of meal wait time distribution, a scatter plot of digital service satisfaction scores, a stacked bar chart showing booking success across student service buildings, and a word cloud built from open ended student feedback. We also built dashboards that combine multiple visualizations to highlight relationships between variables, such as student demographics and infrastructure stress levels. These visual artifacts make it easy to communicate key findings and support our overall data story by showing concretely how students experience different aspects of campus life at Virginia Tech.

A reference section detailing all sources you used.

a. Where did you get your data (provide titles of the pages and links)?
   1) Google Form

b. What references did you use to plan your project; for example, what news articles or magazine articles did you use to help analyze or interpret your data (provide titles of the articles and links)?

   1) "Virginia Tech Sustainability Annual Report 2024-2025: This report highlights student-led "Green RFPs" (Requests for Proposals) and improvements to campus water and energy use. (Sustainability initiatives yield educational and economic returns | Virginia Tech News:
   2) Board of Visitors: Residential & Athletic Infrastructure Planning (Feb 2026): This provides data on the ongoing renovation of aging residence halls (Slusher, Hoge, and Pritchard) and energy code updates.(Board of Visitors advances planning for athletics and residential facilities | Virginia Tech News)
   3) Virginia Tech IT Strategic Plan (2025) This plan outlines the university's "user-centered service" and technology roadmap.(Introducing Virginia Tech's new IT Strategic Plan | VT Division of IT)

c. What data visualizations did you use for inspiration (provide the links for the visualizations)?
   - https://public.tableau.com/app/profile/taofeek.oladigbolu4026/viz/Superstore_Dashboard_17726338267000/cust_dashboard
   - https://public.tableau.com/app/profile/chris.meardon/viz/EmployeeTimeTracker/IndividualConsultantReport

d. What references did you use to help interpret, explain, or provide recommendations for your findings (provide titles and links)?

Ans) We referred to several sources to help interpret our findings and support our recommendations about campus transportation, dining services, and digital systems. These references provided context about student satisfaction, campus mobility, and best practices in university services.

- Virginia Tech Transportation Services – https://parking.vt.edu/
  This source helped us understand transportation options available to students and provided context for analyzing student feedback on commuting and parking.
- Blacksburg Transit (BT Transit) – https://ridebt.org/
  This website provided information about bus routes, schedules, and services, which helped us interpret survey responses related to student satisfaction with campus transportation.
- Virginia Tech Dining Services – https://dining.vt.edu/
  This source helped us understand dining hall operations and services, which was useful when analyzing student feedback on wait times and dining satisfaction.
- Virginia Tech IT Services (Canvas, Hokie SPA, eduroam) – https://it.vt.edu/
  This site provided information on the digital platforms used by students, helping us better explain responses related to campus online systems.